



Top 10 Spring Boot Projects To Destroy The Competition

In this blog, we'll dive into a curated list of Spring Boot projects tailored for different experience levels: beginners, college students (medium-level), and working professionals (advanced).



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If you want to stand out in today's competitive job market, small CRUD apps or basic clones won't cut it anymore. Recruiters can instantly spot projects that were built with one or two AI prompts on Lovabale or Bolt. If you really wish to **destroy the competition**, you need to build **big projects that solve real problems**, designed as if they could be deployed at a startup level. Think beyond "just projects"—focus on building **products** that demonstrate mastery of **Spring Boot** and integration with a full industry-ready tech stack.

If you feel like you already know these concepts but are unsure about which project to build and how to approach it, you’ve come to the right place. In this blog, I’ve shared 10 backend project ideas to help you decide which one to start with. I recommend building at least one intermediate project and one advanced project to make your resume stand out.

Beginner-Level Projects Tech-stack:

Component	Tech/Tools
Database & Persistence	Spring Data JPA, Hibernate JPQL, Query Hints, Indexes
Transactions	@Transactional , Optimistic & Pessimistic Locking
Security & Auth	Spring Security, JWT, OAuth2 Role-Based Access Control
API Design	CRUD APIs, Pagination & Sorting (Pageable), Validation (@Valid), Auditing
Testing & Quality	JUnit 5, Mockito, Spring Boot Test Aim for 85%+ coverage
Logging & Monitoring	SLF4J, Logback
Deployment & Cloud Basics	File Uploads (AWS S3, GCP Storage) Deploy on AWS EC2/Elastic Beanstalk Intro to Docker & CI/CD

Project 1: Blog Management System

Level: Beginner

Create a simple blogging platform where users can create, read, update, and delete blog posts. Include user authentication for posting and commenting.

What You Will Learn: Fundamentals of RESTful APIs, database persistence with JPA, basic authentication, and API documentation. You'll understand how to structure a monolithic application and handle data validation.

Component	Tool	Why It Matters
Database ORM	Spring Data JPA	Simplifies database interactions by providing repository interfaces, reducing boilerplate code for CRUD operations.
API Documentation	Swagger UI	Automatically generates interactive API docs, making it easier for developers (and recruiters) to test and understand your endpoints.
Security	Spring Security	Handles authentication and authorization, teaching secure practices essential for any web app to prevent unauthorized access.

Key Project Features:

- User registration and login.
- CRUD operations for blog posts and comments.
- Role-based access (e.g., admin can delete posts).

Project Challenges:

- Managing one-to-many relationships between entities (e.g., posts and comments) in JPA, which can lead to N+1 query problems if not optimized with fetch types.
- Implementing secure password hashing with BCrypt and JWT tokens for stateless authentication, requiring careful handling of token expiration and refresh mechanisms.
- Handling custom exceptions with `@ControllerAdvice` to provide meaningful error responses without exposing stack traces.

“Similar to personal blogs or content management systems like WordPress, where users create and manage content securely.”

Extra Mile	Tool and Tech	Impact on Recruiters
Add email notifications for new comments using Spring Mail.	Spring Boot Mail Starter, SMTP server (e.g., Gmail).	Shows initiative in integrating external services, highlighting full-stack potential.

Extra Mile	Tool and Tech	Impact on Recruiters
Deploy to Heroku or AWS with a simple CI/CD pipeline.	GitHub Actions, Docker.	Demonstrates deployment knowledge, making you stand out as deployment-ready.
Implement unit tests with JUnit and Mockito.	JUnit 5, Mockito, Spring Boot Test.	Proves testing skills, a key requirement in professional environments.

Add in Resume:

- Developed a secure Blog Management System using Spring Boot, handling 500+ simulated daily posts with 99% uptime; integrated Spring Data JPA and Swagger UI for efficient CRUD and API docs, reducing development time by 30%.
- Implemented role-based authentication with Spring Security and JWT, supporting 1,000 concurrent users; added email notifications via Spring Mail, enhancing user engagement by 40% in tests.
- Wrote comprehensive unit tests covering 85% code, using JUnit and Mockito, ensuring robust error handling and impressing with production-ready code quality.

Project 2: Student Management System

Level: Beginner

Build an application to manage student records, including enrollment, grades, and courses.

What You Will Learn: Entity mapping, query optimization with JPA, and basic API design patterns.

Component	Tool	Why It Matters
Database ORM	Spring Data JPA	Enables efficient querying and pagination for large datasets like student lists.

Component	Tool	Why It Matters
API Documentation	Swagger UI	Provides a user-friendly interface to explore APIs, useful for educational demos.
Security	Spring Security	Secures sensitive data like grades, teaching role-based access control (e.g., teacher vs. student).

Key Project Features:

- CRUD for students, courses, and enrollments.
- Search functionality by name or ID.
- Basic reporting (e.g., average grades).

Project Challenges:

- Handling many-to-many relationships (students to courses) with JPA's **@ManyToMany** annotation, avoiding duplicate entries and optimizing joins to prevent performance bottlenecks.
- Ensuring data integrity during updates using **@Transactional** annotations and optimistic locking with **@Version** to handle concurrent modifications.
- Debugging JPA lazy loading issues with Hibernate's fetch strategies, which can cause **LazyInitializationException** If not managed in service layers.

“Used in educational institutions for tracking student progress, similar to university portals.”

Extra Mile	Tool and Tech	Impact on Recruiters
Integrate with a frontend like React or Angular	React JS, Angular	Shows ability to build end-to-end applications, appealing to full-stack roles.
Add file uploads for student photos using Spring	Spring Web MultipartFile, AWS S3 (optional).	Demonstrates handling binary data, a common real-world feature.

Extra Mile

Tool and Tech

Impact on Recruiters

Multipart.

Add in Resume:

- Engineered a Student Management System in Spring Boot, managing 10,000+ records with pagination; leveraged Spring Data JPA and Swagger for optimized queries, cutting load times by 50%.
- Secured sensitive data with Spring Security's role-based access, handling 500 users; incorporated file uploads via Multipart, boosting data richness and user interaction.
- Applied Lombok for cleaner code, achieving 90% reduction in boilerplate; tested with JUnit, ensuring scalability and reliability for educational platforms.

Project 3: Hospital Management System

Level: Beginner

Develop a system to manage patient records, appointments, and doctor schedules.

What You Will Learn: Transactional operations, validation, and integrating multiple entities.

Component	Tool	Why It Matters
Database ORM	Spring Data JPA	Manages complex entities like patients and appointments with ease.
API Documentation	Swagger UI	Allows quick testing of health-related APIs in a demo.
Security	Spring Security	Protects sensitive health data, complying with privacy standards.

Key Project Features:

- Appointment booking and cancellation.
- Patient history tracking.
- Doctor availability checks.

Project Challenges:

- Ensuring atomic transactions for bookings using Spring's **@Transactional** with propagation levels, rolling back on exceptions.
- Validating input data (e.g., dates with **@Valid** and custom validators) to prevent invalid states, integrating with Bean Validation API for constraints like **@Future**.
- Scaling for multiple users by implementing pessimistic locking in JPA queries to avoid overbooking scenarios in high-concurrency environments.

Extra Mile	Tool and Tech	Impact on Recruiters
Add SMS notifications via Twilio API.	Twilio SDK, Spring Integration.	Integrates third-party services, showing real-world integration skills.
Implement pagination and sorting in queries.	Spring Data Pageable, Criteria API.	Optimizes performance, key for data-heavy apps.
Use Docker for containerization.	Docker Compose, Dockerfile.	Introduces DevOps basics, highly valued in modern teams.

Add in Resume:

- Built a Hospital Management System with Spring Boot, processing 1,000+ appointments daily; utilised Spring Data JPA and Swagger for seamless data handling, improving query efficiency by 40%.
- Enforced data privacy with Spring Security and transactional integrity, supporting 200 concurrent sessions; added SMS alerts via Twilio, enhancing patient satisfaction scores.

- Containerized app with Docker, enabling easy deployment; achieved 95% test coverage with JUnit.

Intermediate Level Projects Tech-stack:

Component	Tech/Tools
System Design & Architecture	Dynamic Pricing Engine (Strategy Pattern) Matchmaking System (OOP + Design Patterns)
Database & Caching	Redis, Caffeine for faster retrieval
Scheduling & Jobs	Spring Scheduler, Quartz for Cron Jobs & Background Tasks
Third-Party Integrations	Weather API, Currency API, Stripe/PayPal, Maps API, Invoicing & Logging APIs
Resilience & Error Handling	Resilience4j, Hystrix (legacy) Retry, Circuit Breakers, Timeouts
Testing & Load Simulation	Apache JMeter, Loader.io, k6 Metrics with Prometheus + Grafana
Deployment & Containers	Docker, Docker Compose Deploy on AWS ECS, GCP Cloud Run

Project 4: Zomato Clone (Food Delivery App)

Level: Intermediate

Replicate a food ordering system with restaurant listings, orders, and admin controls.

What You Will Learn: Caching for performance, transaction management for orders, design patterns (e.g., Singleton for configs), logging/auditing, cron jobs for promotions, and payment integration.

Component	Tool	Why It Matters
Caching	Redis/Spring Cache	Reduces database hits for frequent queries like menu items, improving response times.
Transactions	Spring Transaction Management	Ensures atomicity in order placements and payments.
Logging/Auditing	SLF4J with Logback	Tracks user actions for debugging and compliance.
Scheduling	Spring Scheduler	Automates tasks like daily reports or expirations.
Payments	Stripe API	Handles secure payments, a critical e-commerce feature.

Key Project Features:

- User/restaurant/admin APIs.
- Order tracking with real-time updates.
- Promo code application via cron.

Project Challenges:

- Managing concurrent orders with Redis distributed locks to prevent race conditions in inventory updates.
- Integrating external payment gateways like Stripe, handling webhooks for asynchronous callbacks and idempotency keys for retry safety.

“Food delivery platforms handling high-volume orders daily.”

Extra Mile	Tool and Tech	Impact on Recruiters
Add geolocation with Google Maps API.	Google Maps SDK, Spring WebClient.	Enhances realism, showing spatial data skills.

Extra Mile	Tool and Tech	Impact on Recruiters
Implement rate limiting with Spring.	Spring Boot Bucket4j, Redis.	Prevents abuse, demonstrating security awareness.
Use Elasticsearch for search.	Elasticsearch Starter, Spring Data Elasticsearch.	Optimizes queries, key for scalable apps.

Add in Resume:

- Created Zomato Clone using Spring Boot, processing 5,000+ orders/hour in simulations; integrated Redis caching and Stripe payments, slashing response times by 60%.
- Applied design patterns and Spring Scheduler for automated promotions; added geolocation via Google Maps, improving delivery accuracy in tests.
- Ensured logging with SLF4J and rate limiting, handling 1,000 users; achieved 80% faster searches with Elasticsearch, showcasing scalable e-commerce expertise.

Project 5: Uber Clone (Ride-Sharing App)

Level: Intermediate

Create a ride-booking system with driver matching, fares, and admin dashboards.

What You Will Learn: System design patterns (e.g., Observer for notifications), architectural patterns like layered architecture, caching for locations, and transactions for bookings.

Component	Tool	Why It Matters
Caching	Redis	Caches driver locations for quick matching.
Transactions	Spring Transaction Management	Manages fare calculations and payments atomically.
Logging/Auditing	Log4j and Hibernate envers	Audits ride histories for safety and disputes.

Component	Tool	Why It Matters
Scheduling	Quartz Scheduler	Schedules maintenance notifications.
Payments	PayPal, Stripe, Razorpay API	Secures transactions in a trust-sensitive app.

Key Project Features:

- Real-time ride requests.
- Admin APIs for driver verification.
- Dynamic fare estimation based on factors like distance, time, ratings, etc.

Project Challenges:

- Handling geospatial data with PostGIS and spatial queries, optimizing for nearest driver searches.
- Ensuring transaction isolation levels (e.g., **SERIALIZABLE**) to prevent phantom reads during fare updates in distributed environments.
- Scaling for peak hours by implementing circuit breakers to manage service failures and avoid cascading errors.

Extra Mile	Tool and Tech	Impact on Recruiters
Integrate WebSockets for live tracking.	Spring WebSocket, STOMP.	Adds real-time capabilities, highly sought after.
Use circuit breakers with Resilience4j.	Resilience4j, Spring Cloud.	Shows fault-tolerance knowledge.
Add ML for surge pricing simulation.	Spring AI, TensorFlow Lite.	Introduces AI, broadening your skill set.

Add in Resume:

- Developed Uber Clone in Spring Boot, matching 2,000+ rides/minute; used Redis for location data and PayPal for secure payments, reducing latency by 45%.
- Implemented Observer pattern and Quartz for notifications; integrated WebSockets for real-time tracking, elevating user experience.
- Added Resilience4j for fault tolerance and ML surge pricing, supporting 500 drivers; impressed with 90% uptime in load tests.

Project 6: Airbnb Clone (Vacation Rental App)

Level: Intermediate

Build a listing platform for properties, bookings, and host management.

What You Will Learn: Patterns like Factory for object creation, caching for listings, and auditing for bookings.

Component	Tool	Why It Matters
Caching	Caffeine, Redis	Speeds up property searches.
Transactions	Spring Transaction Management	Handles booking conflicts.
Logging/Auditing	SLF4J and Hibernate envers	Logs user interactions for reviews.
Scheduling	Spring Task, Schedulers	Cleans up expired listings.
Payments	Stripe	Processes secure deposits.

Key Project Features:

- Property search and booking.
- Host admin panels.
- Review system.

Project Challenges:

- Syncing calendars for availability using **iCalendar** format or custom cron jobs, preventing double bookings with optimistic concurrency control.
- Managing multi-currency payments with exchange rate APIs, ensuring precision with BigDecimal to avoid floating-point errors.
- Maintaining data consistency across users via distributed caching invalidation strategies.

Extra Mile	Tool and Tech	Impact on Recruiters
Add image uploads with AWS S3.	AWS SDK, Spring Cloud AWS.	Handles cloud storage, modern tech stack.
Implement i18n for multi-language.	Spring Messages, ResourceBundle.	Shows global app design.
Use GraphQL for flexible APIs.	Spring GraphQL, Apollo.	Advanced API tech, impressive for mid-level.

Add in Resume:

- Engineered Airbnb Clone with Spring Boot, listing 10,000+ properties; employed Caffeine caching and Stripe for bookings, boosting search speed by 55%.
- Used Factory pattern and Spring Task for maintenance; added AWS S3 image uploads, enriching listings with multimedia.
- Supported i18n and GraphQL for flexible queries, handling global users; achieved 85% faster API responses, highlighting international scalability.

Project 7: HDFC Net Banking (Banking Application)

Level: Intermediate

Develop a basic banking system with accounts, transfers, and admin oversight.

What You Will Learn: Secure transactions, auditing for compliance, schedulers for interest calculations.

Component	Tool	Why It Matters
Caching	Redis	Caches account balances for quick access.
Transactions	Spring Transaction Management	Ensures ACID properties in transfers.
Logging/Auditing	Logback, Slf4j, Hibernate envers	Records all financial activities for audits.
Scheduling	Cron Jobs in Spring	Automates monthly statements.
Payments	PayPal/Stripe	Simulates fund transfers securely.

Key Project Features:

- Account creation and transfers.
- Transaction history.
- Admin fraud detection.

Project Challenges:

- Implementing high-security with encryption (e.g., AES for data at rest) and secure random number generation for tokens.
- Handling decimal precision in monetary values using Money API or BigDecimal, avoiding rounding errors in calculations.
- Ensuring compliance by integrating audit logs with timestamps and immutable records using AspectJ for AOP.

“Online banking apps for daily finance management.”

Extra Mile	Tool and Tech	Impact on Recruiters
Add two-factor auth with Spring.	Spring Security OAuth	Enhances security, critical for fintech.
Integrate with external APIs for currency exchange.	Spring WebClient.	Shows third-party integrations.
Use blockchain simulation for ledgers (Just a concept)	Hyperledger Fabric (simulated), Spring Integration.	Futuristic touch, stands out in interviews.

Add in Resume:

- Built Banking Application in Spring Boot, managing \$1M simulated transactions; utilized Redis caching and Spring Transactions for ACID compliance, minimizing errors to 0.1%.
- Incorporated Logback auditing and cron jobs for statements; added 2FA with Spring Security, strengthening fraud prevention.
- Integrated currency APIs and blockchain sim, supporting multi-currency; delivered 95% audit accuracy, positioning as fintech innovator.

Advanced Level Projects Tech-stack:

Component	Tech/Tools
Microservices Architecture	Spring Boot + Spring Cloud (Eureka, Config Server, API Gateway) Authentication Service (Keycloak, OAuth2.0, JWT) Notification & Media Management Services
Polyglot Persistence	PostgreSQL/MySQL (ACID) DynamoDB/Cassandra (scaling) Redis Cluster (distributed caching) Elasticsearch (search & aggregation) PostGIS/MongoDB Atlas (geospatial queries) Neo4j (graph DB) Pinecone/Weaviate/Milvus (vector DB for ML)
Messaging & Events	Kafka, RabbitMQ Event Sourcing & CQRS Handling surge traffic

Component	Tech/Tools
Distributed Transactions	Saga Pattern, Two-Phase Commit, Idempotency
Inter-Service Communication	REST (Feign) gRPC WebSockets Kafka Messaging (async)
Observability & Monitoring	Zipkin, Jaeger (tracing) Prometheus + Grafana (metrics) ELK Stack (logging) Correlation IDs
Deployment & Cloud-Native	Kubernetes, Helm, Jenkins/GitHub Actions (CI/CD) AWS Lambda, Cloud Functions (serverless) Terraform, CloudFormation (IaC)
Security & Compliance	Secrets Management (Vault, AWS Secrets Manager) GDPR/PCI compliance
Scalability & Advanced Features	Sharding & Replication Multi-Tenancy Architecture (SaaS) AI/ML Integration via REST/gRPC API Gateway with Rate Limiting & Throttling

Project 8: BookMyShow Clone (Ticket Booking System)

Level: Advanced

Build a microservices architecture for movie/event ticketing, with services for users, bookings, payments, and notifications.

What You Will Learn: Microservices design, Kubernetes deployment, Kafka for messaging, observability with Prometheus/Grafana, logging (ELK stack), auditing, and inter-service communication via REST/gRPC.

Key Project Features:



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- Observability dashboards.

Component	Tool	Why It Matters
Orchestration	Kubernetes	Automates deployment and scaling for high availability.
Messaging	Kafka	Enables asynchronous communication for notifications and events.
Observability	Prometheus + Grafana	Monitors metrics and alerts for proactive issue resolution.
Logging/Auditing	ELK Stack (Elasticsearch, Logstash, Kibana)	Centralizes logs for debugging across services.
Inter-Service	Spring Cloud Gateway	Routes and secures communication between microservices.

Project Challenges:

- Scaling to billions of transactions by configuring Kafka partitions and consumer groups for high throughput, avoiding bottlenecks in topic replication.
- Managing service discovery in Kubernetes with tools like Consul or Eureka, handling pod evictions and rolling updates without service interruption.
- Achieving eventual consistency in distributed systems using the Saga pattern or 2PC, mitigating failures with compensating transactions.

“High-traffic ticketing platforms during peak events like concerts, This level of project shows your backend prowess.”

Extra Mile	Tool and Tech	Impact on Recruiters
Implement CI/CD with Jenkins on Kubernetes.	Jenkins, Helm Charts.	Shows DevOps expertise, essential for senior roles.

Extra Mile	Tool and Tech	Impact on Recruiters
Add AI for recommendation engines.	Spring AI, Apache Mahout.	Combines backend with ML, highly impressive.
Simulate load testing with JMeter.	Apache JMeter, Locust.	Proves scalability claims with data.

Add in Resume:

- Architected BookMyShow Clone as microservices in Spring Boot, scaling to 1M users and 1B transactions; deployed on Kubernetes with Kafka messaging, achieving 99.9% availability.
- Integrated Prometheus/Grafana for observability and ELK for logging; added CI/CD via Jenkins, accelerating deployments by 70%.
- Implemented AI recommendations and JMeter testing; reduced latency by 50%, demonstrating enterprise-grade resilience.

Project 9: Telegram Clone (WebSockets Chat Application)

Develop a real-time chat app as microservices, with user auth, messaging, and group chats, deployed on Kubernetes.

What You Will Learn: WebSockets for real-time, Kafka for message queuing, observability, logging, and inter-service sync.

Key Project Features:

- Scalable to millions of concurrent users.
- Message persistence and delivery guarantees.
- Monitoring for chat latency.
- Audit user interactions.

Component	Tool	Why It Matters
Orchestration	Kubernetes	Manages WebSocket pods for horizontal scaling.
Messaging	Kafka	Buffers and distributes messages reliably.
Observability	Micrometer + Prometheus	Tracks connection metrics.
Logging/Auditing	ELK, Hibernate Envers	Logs chat histories securely.
Inter-Service	WebSockets with STOMP	Enables bidirectional real-time communication.

Project Challenges:

- Handling WebSocket connections at scale by configuring session timeouts and heartbeat pings to detect disconnections.
- Partitioning Kafka topics efficiently for group chats, balancing load across brokers to maintain low latency.
- Ensuring data privacy with end-to-end encryption using libs like Bouncy Castle, complying with GDPR-like standards.

“Messaging apps like Slack for team communication.”

Extra Mile	Tool and Tech	Impact on Recruiters
Add end-to-end encryption.	Bouncy Castle, Spring Security.	Security focus, vital for privacy apps.
Integrate with push notifications (FCM).	Firebase SDK, Spring Integration.	Enhances user experience.
Use serverless functions for bots.	AWS Lambda, Spring Cloud Function.	Modern architecture blend.

Add in Resume:

- Built a WebSockets Chat App in Spring Boot microservices, supporting 1M users; integrated Kafka and Kubernetes for real-time messaging, with 99% delivery rate.
- Added Micrometer/Prometheus monitoring and ELK logging; implemented E2E encryption, prioritizing data privacy.
- Incorporated FCM notifications and serverless bots; reduced latency by 40%, highlighting scalable communication tech.

Project 10: Social Media App like LinkedIn

Level: Advanced

Build a professional networking platform with microservices for profiles, posts, connections, and jobs.

What You Will Learn: Scaling for billions of interactions, Kafka for feeds, Kubernetes deployment, full observability stack.

Key Project Features:

- Feed generation with Kafka streams.
- Inter-service calls for recommendations.
- Logging all user actions.
- Handle a massive user base.

Component	Tool	Why It Matters
Orchestration	Kubernetes	Orchestrates services for global users.
Messaging	Kafka	Processes post events and notifications.
Observability	Zipkin for tracing	Visualizes service interactions.
Logging/Auditing	ELK Stack	Audits for content moderation.

Component	Tool	Why It Matters
Inter-Service	Spring Cloud Feign	Simplifies client-side calls.

Project Challenges:

- Personalizing feeds at scale using Kafka Streams for real-time processing, managing state stores for user preferences.
- Managing graph data for connections with eventual consistency, using CRDTs or distributed graphs to resolve conflicts.
- Implementing distributed caching with Redis clusters to handle cache invalidation across microservices.

Extra Mile	Tool and Tech	Impact on Recruiters
Add graph DB like Neo4j for connections.	Neo4j Driver, Spring Data Neo4j.	Handles complex relationships efficiently.
Implement content moderation with AI.	Spring AI, Hugging Face models.	Addresses real-world challenges.
Deploy on multi-cloud (AWS + GCP).	Terraform, Kubernetes Federation.	Shows cloud-agnostic skills.

Add in Resume:

- Created LinkedIn-like Social App with Spring Boot microservices, for 10M users and 1B interactions; used Kafka Streams and Kubernetes, optimizing feeds by 65%.
- Integrated Zipkin tracing and ELK auditing; added Neo4j for connections, enabling efficient networking graphs.
- Applied AI moderation and multi-cloud deploy; improved engagement by 50%, proving enterprise social platform mastery.

By completing these projects, you'll not only master Spring Boot but also gain insights into real-world development. Tailor them to your resume, host on GitHub, and explain your choices in interviews.

Tech Stack Summary:

Component	Beginner Level	Intermediate Level	Advanced Level
Database & Persistence	Relational DB Mappings (Spring Data JPA, Hibernate) Query Optimization (JPQL, Indexes)	Caching with Redis/Caffeine Dynamic Pricing with DB-driven rules	Polyglot Persistence: Postgres (ACID), DynamoDB/Cassandra (scaling), Redis Cluster (distributed caching), Elasticsearch (search), PostGIS (geospatial), Neo4j (graph), Vector DB (ML)
Transactions	@Transactional Optimistic & Pessimistic Locking	Handling concurrent transactions in pricing & booking systems	Distributed Transactions (Saga, Two-Phase Commit, Idempotency)
Security & Auth	Spring Security (JWT, OAuth2) Role-Based Access Control	API Security for Payments & Third-Party APIs	Enterprise Auth with OAuth2.0, Keycloak, Secrets Management (Vault, AWS Secrets Manager)
API & Communication	CRUD APIs Pagination, Validation, Auditing	Third-Party Integrations (Weather, Currency, Payments, Maps)	Inter-Service Communication: REST (Feign), gRPC, WebSockets, Kafka Messaging
Testing & Quality	Unit & Integration Tests (JUnit, Mockito) 85%+ coverage	Load Testing with JMeter/Loader.io, Stress Testing with k6	Chaos Testing, CI/CD Test Automation (Jenkins, GitHub Actions)
Scheduling & Jobs	Basic Scheduling with Spring Scheduler	Cron Jobs & Background Processing (Quartz)	Distributed Schedulers in Microservices, Event-Driven Scheduling (Kafka Streams)

Component	Beginner Level	Intermediate Level	Advanced Level
Observability & Logging	Logging (SLF4J, Logback) Basic Auditing	Monitoring performance under stress tests with Grafana + Prometheus	Distributed Tracing (Zipkin, Jaeger), Centralized Logging (ELK Stack), Correlation IDs
Messaging & Events	—	Retry & Resilience with Resilience4j, Circuit Breakers	Kafka, RabbitMQ for async processing, Event Sourcing & CQRS for large-scale systems
Deployment & Cloud	File Uploads to AWS S3/GCP Deploy on AWS EC2/Elastic Beanstalk Intro to Docker	Docker & Docker Compose Deploy on AWS ECS/GCP Cloud Run	Kubernetes (K8s), Helm, Serverless (AWS Lambda, Cloud Functions) Infra as Code (Terraform)
Architecture & Design	Basic CRUD Architecture	Matchmaking Systems (OOP + Design Patterns)	Microservices Ecosystem (Auth, Config, Gateway, Notifications, Domain Services) DDD + Multi-Tenancy Sharding, SaaS Architecture
Advanced Features	—	—	AI/ML Integration (TensorFlow Serving, PyTorch + gRPC) Vector DBs for semantic search Rate Limiting & Throttling (API Gateway)

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